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1 Summary

Definition 1.1. We are discussing a Bond with the following terms:

- A = *par value*, the amount paid at maturity;
- n = number of compounding periods per year;
- T is the number of years *remaining* to maturity;
  - $N := nT$ is the *total* number of compounding periods.
- Y is the annual "desired/prevailing[1]" rate of return (a.k.a. the *yield to maturity*)
  - $y := Y/n$ is the per-period rate of return.
- R is the *annual* coupon rate;
  - $\rho := R/n$ is the *per-period* coupon rate.

Finally, we let $P := P(\rho, y, N)$ be the *current* price of the bond.

We may derive (see below) that

$$P(\rho, y, N) = A + A\big(\tfrac{\rho}{y} - 1\big)\big(1 - (1+y)^{-N}\big). \tag{1.1}$$

1.1 **Derivation.** If $y = Y/n$ is the "prevailing" per-period rate for the class of Bond, then the per-period *discount factor* is

$$\nu := (1 + Y/n) = (1 + y).$$

With this, the present value of the coupon payments (each of size $A\rho$) and the present value of the face value are

$$\text{PV(coupons)} + \text{PV(Par)} = A\rho(\nu^{-1} + \cdots + \nu^{-N}) + A\nu^{-N} = A\rho \cdot \frac{1 - \nu^{-N}}{\nu(1 - \nu^{-1})} + A\nu^{-N}.$$

Simplifying and using the fact that $\nu - 1 = y$, we arrive at the formula

$$P = A + A\Big(\frac{\rho}{y} - 1\Big)\big(1 - (1+y)^{-N}\big)$$

1.2 **Extra/Side Notes.**

(1) In the context of Bonds, the coupon rate is taken to be constant for the life of the Bond.

[1] We can assume this is the *risk-free rate*.